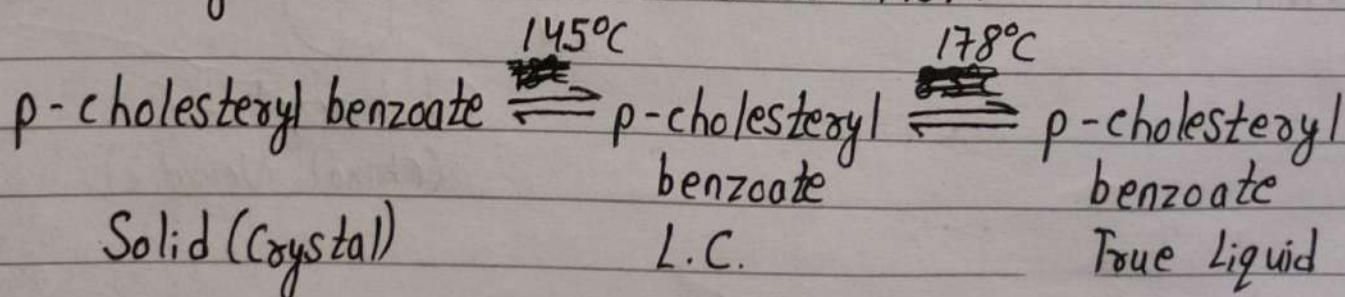


Module - I: Liquid Crystal

Liquid Crystal

- Liquid crystal are state of matter that have properties between solid (crystal) and liquid.
- They flow like liquid but shows some long range molecular order like crystal.
- On heating certain solids undergoes sharp-phase changes one after another. The translucent and turbid liquid exhibit n-isotropy i.e. they have different physical properties in different directions, on the other hand true liquids are isotropy.
- The substances shows, behaviour of liquid crystal usually long chain organic molecules and the terminal group are $-OR$, $-COOR$, etc.
- In 1991, P.G. de Gennes got a novel ~~prize~~^{prize} in physics for his contribution in liquid crystal and polymer.
- The first liquid crystal discovered in 1889 was cholesteryl benzoate ($C_{27}H_{45}COOC_6H_5$).



→ Some common examples of liquid crystals are;

p-azoxy anisol

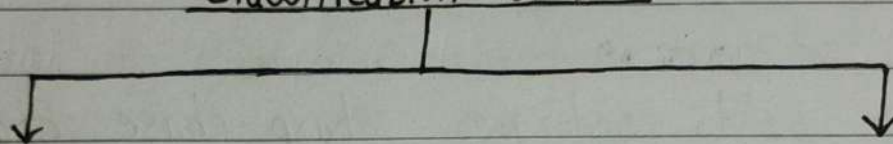
p-azoxy phenitol

p-methoxy cinnamic acid

diethyl benzidine.

Classification of Liquid Crystal (L.C.)

Classification of L.C.



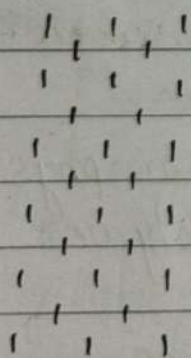
Thermotropic L.C.

[Formed by changed in temperature]

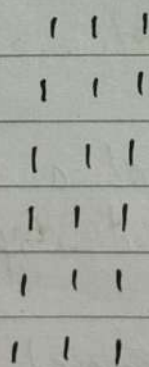
Types: Nematic L.C.

Smectic L.C.

Cholesteryl L.C.



Nematic L.C.



Smectic L.C.

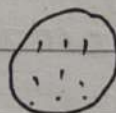
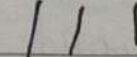
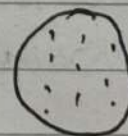
Lyotropic L.C.

[Formed by dissolving in Lyophilic]

[Water Attracting]

Depends of conc. and temp.

Ex: Micelles & Vesicles



Cholesteryl L.C.

(Chiral Nematic)

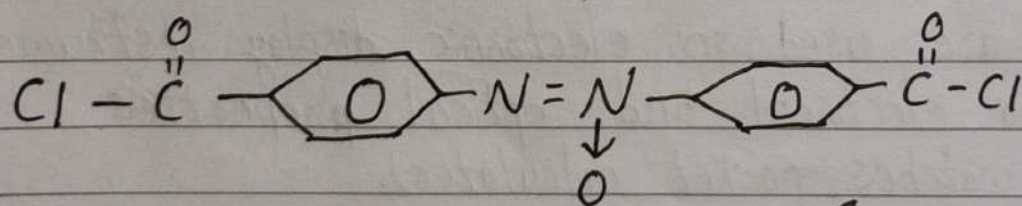
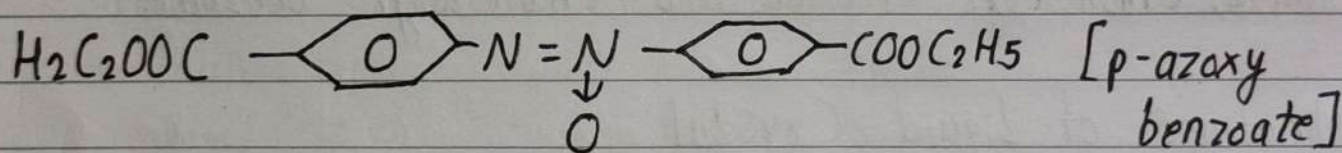
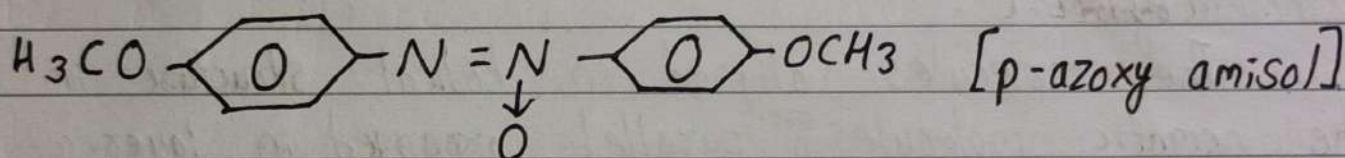
① Nematic L.C.:

The molecules are parallel to each other, the nematic L.C. loses their isotropy and form clear solution, when they are oriented in an electrical or magnetic field.

In nematic L.C., molecules are aligned parallel but not positional order.

NLC are viscous in nature.

For example;



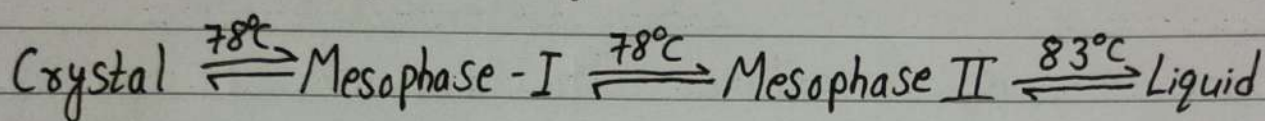
[isotropic acid halide]
[Not L.C.]

② Smectic L.C:

The molecules are arranged in parallel but these are arranged in layers.

The layers can be slide each other. S.L.C do not flow as a normal liquid but they flow in layers.

Example; Cholesteryl mesitylate (Two mesoform)



③ Cholesteral L.C:

The molecules aligned in helical structure in which the nematic molecules are parallel arranged in layers.

Example; Cholesteryl chloride and Cholesteryl benzoate.

Applications of Liquid Crystal

1. Liquid crystals are used in electronic display instruments because of their electrical and optical properties (digital wrist watches, pocket calculator).
2. Liquid crystals are used for detecting tumors in the body by using a method called thermography.
3. Liquid crystals are used as solvents for studying the structure of anisotropic molecules spectroscopically.

4. Liquid crystals are used as commercial lubricants.

5. Liquid crystals are most suited to biological functions.